

# Trigonometric Integrals

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**EXAMPLE 1** Evaluate  $\int \cos^3 x \, dx$ .

**SOLUTION** Simply substituting  $u = \cos x$  isn't helpful, since then  $du = -\sin x \, dx$ . In order to integrate powers of cosine, we would need an extra  $\sin x$  factor. Similarly, a power of sine would require an extra  $\cos x$  factor. Thus, here we can separate one cosine factor and convert the remaining  $\cos^2 x$  factor to an expression involving sine using the identity  $\sin^2 x + \cos^2 x = 1$ :

$$\cos^3 x = \cos^2 x \cdot \cos x = (1 - \sin^2 x) \cos x$$

We can then evaluate the integral by substituting  $u = \sin x$ , so  $du = \cos x \, dx$  and

$$\begin{aligned} \int \cos^3 x \, dx &= \int \cos^2 x \cdot \cos x \, dx = \int (1 - \sin^2 x) \cos x \, dx \\ &= \int (1 - u^2) \, du = u - \frac{1}{3}u^3 + C \\ &= \sin x - \frac{1}{3}\sin^3 x + C \end{aligned}$$

$$\int \cos^5 x \, dx.$$

**Solution**

$$\int \cos^5 x \, dx = \int \cos^4 x \cos x \, dx = \int (1 - \sin^2 x)^2 d(\sin x) \quad \cos x \, dx = d(\sin x)$$

$$= \int (1 - u^2)^2 du \quad u = \sin x$$

$$= \int (1 - 2u^2 + u^4) du \quad \text{Square } 1 - u^2.$$

$$= u - \frac{2}{3}u^3 + \frac{1}{5}u^5 + C = \sin x - \frac{2}{3}\sin^3 x + \frac{1}{5}\sin^5 x + C.$$

$$\int \sin x \left( \cos^4 x - \sin x \right) dx$$

The integration process involves the substitution  $u = \cos x$ ,  $du = -\sin x \, dx$  and the trigonometric identity  $\sin^2 x = \frac{1 - \cos 2x}{2}$  to yield

$$\begin{aligned} \int \sin x \left( \cos^4 x - \sin x \right) dx &= \int \sin x \cos^4 x \, dx - \int \sin^2 x \, dx \\ &= - \int u^4 \, du - \frac{1}{2} \int (1 - \cos 2x) \, dx \\ &= -\frac{u^5}{5} - \frac{1}{2}x + \frac{1}{4} \int 2 \cos 2x \, dx \\ &= -\frac{\cos^5 x}{5} - \frac{1}{2}x + \frac{1}{4} \sin 2x + C. \end{aligned}$$

**EXAMPLE 2** Find  $\int \sin^5 x \cos^2 x \, dx$

**SOLUTION** We could convert  $\cos^2 x$  to  $1 - \sin^2 x$ , but we would be left with an expression in terms of  $\sin x$  with no extra  $\cos x$  factor. Instead, we separate a single sine factor and rewrite the remaining  $\sin^4 x$  factor in terms of  $\cos x$ :

$$\sin^5 x \cos^2 x = (\sin^2 x)^2 \cos^2 x \sin x = (1 - \cos^2 x)^2 \cos^2 x \sin x$$

Substituting  $u = \cos x$ , we have  $du = -\sin x \, dx$  and so

$$\begin{aligned} \int \sin^5 x \cos^2 x \, dx &= \int (\sin^2 x)^2 \cos^2 x \sin x \, dx \\ &= \int (1 - \cos^2 x)^2 \cos^2 x \sin x \, dx \\ &= \int (1 - u^2)^2 u^2 (-du) = -\int (u^2 - 2u^4 + u^6) \, du \\ &= -\left(\frac{u^3}{3} - 2\frac{u^5}{5} + \frac{u^7}{7}\right) + C \\ &= -\frac{1}{3}\cos^3 x + \frac{2}{5}\cos^5 x - \frac{1}{7}\cos^7 x + C \end{aligned}$$

In the preceding examples, an odd power of sine or cosine enabled us to separate a single factor and convert the remaining even power. If the integrand contains even powers of both sine and cosine, this strategy fails. In this case,

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x) \quad \text{and} \quad \cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

**EXAMPLE** Find  $\int \sin^4 x \, dx$ .

We could evaluate this integral using the reduction formula for  $\int \sin^n x \, dx$

write  $\sin^4 x = (\sin^2 x)^2$  and use a half-angle formula:

$$\begin{aligned} \int \sin^4 x \, dx &= \int (\sin^2 x)^2 \, dx \\ &= \int \left( \frac{1 - \cos 2x}{2} \right)^2 \, dx \\ &= \frac{1}{4} \int (1 - 2 \cos 2x + \cos^2 2x) \, dx \end{aligned}$$

Since  $\cos^2 2x$  occurs, we must use another half-angle formula

$$\cos^2 2x = \frac{1}{2}(1 + \cos 4x)$$

$$\begin{aligned}
 \int \sin^4 x \, dx &= \frac{1}{4} \int [1 - 2 \cos 2x + \frac{1}{2}(1 + \cos 4x)] \, dx \\
 &= \frac{1}{4} \int (\frac{3}{2} - 2 \cos 2x + \frac{1}{2} \cos 4x) \, dx \\
 &= \frac{1}{4} (\frac{3}{2}x - \sin 2x + \frac{1}{8} \sin 4x) + C
 \end{aligned}$$

**Strategy for Evaluating**  $\int \sin^m x \cos^n x \, dx$

- (a) If the power of cosine is odd ( $n = 2k + 1$ ), save one cosine factor and use  $\cos^2 x = 1 - \sin^2 x$  to express the remaining factors in terms of sine:

$$\begin{aligned}
 \int \sin^m x \cos^{2k+1} x \, dx &= \int \sin^m x (\cos^2 x)^k \cos x \, dx \\
 &= \int \sin^m x (1 - \sin^2 x)^k \cos x \, dx
 \end{aligned}$$

Then substitute  $u = \sin x$ .

- (b) If the power of sine is odd ( $m = 2k + 1$ ), save one sine factor and use  $\sin^2 x = 1 - \cos^2 x$  to express the remaining factors in terms of cosine:

$$\begin{aligned}
 \int \sin^{2k+1} x \cos^n x \, dx &= \int (\sin^2 x)^k \cos^n x \sin x \, dx \\
 &= \int (1 - \cos^2 x)^k \cos^n x \sin x \, dx
 \end{aligned}$$

Then substitute  $u = \cos x$ . [Note that if the powers of both sine and cosine are odd, either (a) or (b) can be used.]

- (c) If the powers of both sine and cosine are even, use the half-angle identities

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x) \qquad \cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

It is sometimes helpful to use the identity

$$\sin x \cos x = \frac{1}{2} \sin 2x$$

$$\int \sin^2 x \cos^4 x \, dx.$$

**Solution**

$$\begin{aligned} \int \sin^2 x \cos^4 x \, dx &= \int \left( \frac{1 - \cos 2x}{2} \right) \left( \frac{1 + \cos 2x}{2} \right)^2 dx && m \text{ and } n \text{ both even} \\ &= \frac{1}{8} \int (1 - \cos 2x)(1 + 2 \cos 2x + \cos^2 2x) \, dx \\ &= \frac{1}{8} \int (1 + \cos 2x - \cos^2 2x - \cos^3 2x) \, dx \\ &= \frac{1}{8} \left[ x + \frac{1}{2} \sin 2x - \int (\cos^2 2x + \cos^3 2x) \, dx \right]. \end{aligned}$$

For the term involving  $\cos^2 2x$ , we use

$$\begin{aligned} \int \cos^2 2x \, dx &= \frac{1}{2} \int (1 + \cos 4x) \, dx \\ &= \frac{1}{2} \left( x + \frac{1}{4} \sin 4x \right). \end{aligned}$$

Omitting the constant of integration until the final result

$$\begin{aligned}
 \int \cos^3 2x \, dx &= \int (1 - \sin^2 2x) \cos 2x \, dx && u = \sin 2x, \\
 &&& du = 2 \cos 2x \, dx \\
 &= \frac{1}{2} \int (1 - u^2) \, du = \frac{1}{2} \left( \sin 2x - \frac{1}{3} \sin^3 2x \right). && \text{Again} \\
 &&& \text{omitting } C
 \end{aligned}$$

Combining everything and simplifying, we get

$$\int \sin^2 x \cos^4 x \, dx = \frac{1}{16} \left( x - \frac{1}{4} \sin 4x + \frac{1}{3} \sin^3 2x \right) + C.$$



Evaluate: (a)  $\int \sin^3 x \cos^8 x \, dx$  and (b)  $\int \cos^5 ax \, dx$ .

**Solution**

$$\begin{aligned} \text{(a)} \quad \int \sin^3 x \cos^8 x \, dx &= \int (1 - \cos^2 x) \cos^8 x \sin x \, dx && \text{Let } u = \cos x, \\ & && du = -\sin x \, dx. \\ &= - \int (1 - u^2) u^8 \, du = \int (u^{10} - u^8) \, du \\ &= \frac{u^{11}}{11} - \frac{u^9}{9} + C = \frac{1}{11} \cos^{11} x - \frac{1}{9} \cos^9 x + C. \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \int \cos^5 ax \, dx &= \int (1 - \sin^2 ax)^2 \cos ax \, dx && \text{Let } u = \sin ax, \\ & && du = a \cos ax \, dx. \\ &= \frac{1}{a} \int (1 - u^2)^2 \, du = \frac{1}{a} \int (1 - 2u^2 + u^4) \, du \\ &= \frac{1}{a} \left( u - \frac{2}{3} u^3 + \frac{1}{5} u^5 \right) + C \\ &= \frac{1}{a} \left( \sin ax - \frac{2}{3} \sin^3 ax + \frac{1}{5} \sin^5 ax \right) + C. \end{aligned}$$

$$\begin{aligned}
\int \cos^2 x \, dx &= \frac{1}{2} \int (1 + \cos 2x) \, dx \\
&= \frac{x}{2} + \frac{1}{4} \sin 2x + C \\
&= \frac{1}{2}(x + \sin x \cos x) + C \quad (\text{since } \sin 2x = 2 \sin x \cos x).
\end{aligned}$$

Evaluate the integral:

$$\int (\cos^4 x - \sin^4 x) dx$$

Solution: We observe that the integrand factors

$$\cos^4 x - \sin^4 x = (\cos^2 x - \sin^2 x)(\cos^2 x + \sin^2 x)$$

Applying the identities

$$\cos^2 x - \sin^2 x = \cos(2x) \quad \text{and} \quad \cos^2 x + \sin^2 x = 1$$

gives us:

$$\int (\cos^4 x - \sin^4 x) dx = \int \cos(2x) dx$$

doing the substitution  $u = 2x$ ,  $du = 2dx$ ,

$$\frac{1}{2} \int \cos(u) du = \frac{1}{2} \sin(u) + C = \frac{1}{2} \sin(2x) + C$$

$$\int \tan x \, dx = \int \frac{\sin x}{\cos x} \, dx \quad \begin{array}{l} \text{Let } u = \cos x. \\ \text{Then } du = -\sin x \, dx. \end{array}$$

$$= - \int \frac{du}{u} = -\ln |u| + C$$

$$= -\ln |\cos x| + C = \ln \left| \frac{1}{\cos x} \right| + C = \ln |\sec x| + C.$$

$$\int \sec^2 x \, dx = \tan x + C.$$

$$\int \tan^4 x \, dx.$$

**Solution**

$$\begin{aligned}\int \tan^4 x \, dx &= \int \tan^2 x \cdot \tan^2 x \, dx = \int \tan^2 x \cdot (\sec^2 x - 1) \, dx \\&= \int \tan^2 x \sec^2 x \, dx - \int \tan^2 x \, dx \\&= \int \tan^2 x \sec^2 x \, dx - \int (\sec^2 x - 1) \, dx \\&= \int \tan^2 x \sec^2 x \, dx - \int \sec^2 x \, dx + \int dx.\end{aligned}$$

In the first integral, we let

$$u = \tan x, \quad du = \sec^2 x \, dx$$

and have

$$\int u^2 \, du = \frac{1}{3} u^3 + C_1.$$

The remaining integrals are standard forms, so

$$\int \tan^4 x \, dx = \frac{1}{3} \tan^3 x - \tan x + x + C.$$

Evaluate

$$\int \sec^3 x \, dx.$$

**Solution** We integrate by parts using

$$u = \sec x, \quad dv = \sec^2 x \, dx, \quad v = \tan x, \quad du = \sec x \tan x \, dx.$$

Then

$$\begin{aligned} \int \sec^3 x \, dx &= \sec x \tan x - \int (\tan x)(\sec x \tan x \, dx) \\ &= \sec x \tan x - \int (\sec^2 x - 1) \sec x \, dx && \tan^2 x = \sec^2 x - 1 \\ &= \sec x \tan x + \int \sec x \, dx - \int \sec^3 x \, dx. \end{aligned}$$

Combining the two secant-cubed integrals gives

$$2 \int \sec^3 x \, dx = \sec x \tan x + \int \sec x \, dx$$

and

$$\int \sec^3 x \, dx = \frac{1}{2} \sec x \tan x + \frac{1}{2} \ln |\sec x + \tan x| + C.$$

We can use a similar strategy to evaluate integrals of the form  $\int \tan^m x \sec^n x \, dx$ . Since  $(d/dx) \tan x = \sec^2 x$ , we can separate a  $\sec^2 x$  factor and convert the remaining (even) power of secant to an expression involving tangent using the identity  $\sec^2 x = 1 + \tan^2 x$ . Or, since  $(d/dx) \sec x = \sec x \tan x$ , we can separate a  $\sec x \tan x$  factor and convert the remaining (even) power of tangent to secant.

**EXAMPLE** Evaluate  $\int \tan^6 x \sec^4 x \, dx$ .

**SOLUTION** If we separate one  $\sec^2 x$  factor, we can express the remaining  $\sec^2 x$  factor in terms of tangent using the identity  $\sec^2 x = 1 + \tan^2 x$ . We can then evaluate the integral by substituting  $u = \tan x$  with  $du = \sec^2 x \, dx$ :

$$\begin{aligned} \int \tan^6 x \sec^4 x \, dx &= \int \tan^6 x \sec^2 x \sec^2 x \, dx \\ &= \int \tan^6 x (1 + \tan^2 x) \sec^2 x \, dx \\ &= \int u^6 (1 + u^2) \, du = \int (u^6 + u^8) \, du \\ &= \frac{u^7}{7} + \frac{u^9}{9} + C \\ &= \frac{1}{7} \tan^7 x + \frac{1}{9} \tan^9 x + C \end{aligned}$$

**EXAMPLE** Find  $\int \tan^5 \theta \sec^7 \theta d\theta$ .

**SOLUTION** If we separate a  $\sec^2 \theta$  factor, as in the preceding example, we are left with a  $\sec^5 \theta$  factor, which isn't easily converted to tangent. However, if we separate a  $\sec \theta \tan \theta$  factor, we can convert the remaining power of tangent to an expression involving only secant using the identity  $\tan^2 \theta = \sec^2 \theta - 1$ . We can then evaluate the integral by substituting  $u = \sec \theta$ , so  $du = \sec \theta \tan \theta d\theta$ :

$$\begin{aligned}\int \tan^5 \theta \sec^7 \theta d\theta &= \int \tan^4 \theta \sec^6 \theta \sec \theta \tan \theta d\theta \\&= \int (\sec^2 \theta - 1)^2 \sec^6 \theta \sec \theta \tan \theta d\theta \\&= \int (u^2 - 1)^2 u^6 du = \int (u^{10} - 2u^8 + u^6) du \\&= \frac{u^{11}}{11} - 2 \frac{u^9}{9} + \frac{u^7}{7} + C \\&= \frac{1}{11} \sec^{11} \theta - \frac{2}{9} \sec^9 \theta + \frac{1}{7} \sec^7 \theta + C\end{aligned}$$



### Strategy for Evaluating $\int \tan^m x \sec^n x dx$

- (a) If the power of secant is even ( $n = 2k, k \geq 2$ ), save a factor of  $\sec^2 x$  and use  $\sec^2 x = 1 + \tan^2 x$  to express the remaining factors in terms of  $\tan x$ :

$$\begin{aligned}\int \tan^m x \sec^{2k} x dx &= \int \tan^m x (\sec^2 x)^{k-1} \sec^2 x dx \\ &= \int \tan^m x (1 + \tan^2 x)^{k-1} \sec^2 x dx\end{aligned}$$

Then substitute  $u = \tan x$ .

- (b) If the power of tangent is odd ( $m = 2k + 1$ ), save a factor of  $\sec x \tan x$  and use  $\tan^2 x = \sec^2 x - 1$  to express the remaining factors in terms of  $\sec x$ :

$$\begin{aligned}\int \tan^{2k+1} x \sec^n x dx &= \int (\tan^2 x)^k \sec^{n-1} x \sec x \tan x dx \\ &= \int (\sec^2 x - 1)^k \sec^{n-1} x \sec x \tan x dx\end{aligned}$$

Then substitute  $u = \sec x$ .

$$\int \tan x \, dx = \ln |\sec x| + C$$

$$\int \sec x \, dx = \ln |\sec x + \tan x| + C$$

$$\begin{aligned} \int \sec x \, dx &= \int \sec x \frac{\sec x + \tan x}{\sec x + \tan x} \, dx \\ &= \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} \, dx \end{aligned}$$

If we substitute  $u = \sec x + \tan x$ , then  $du = (\sec x \tan x + \sec^2 x) \, dx$ , so the integral becomes  $\int (1/u) \, du = \ln |u| + C$ . Thus, we have

$$\int \sec x \, dx = \ln |\sec x + \tan x| + C$$

**EXAMPLE** Find  $\int \tan^3 x \, dx$ .

**SOLUTION** Here only  $\tan x$  occurs, so we use  $\tan^2 x = \sec^2 x - 1$  to rewrite a  $\tan^2 x$  factor in terms of  $\sec^2 x$ :

$$\begin{aligned}\int \tan^3 x \, dx &= \int \tan x \tan^2 x \, dx \\&= \int \tan x (\sec^2 x - 1) \, dx \\&= \int \tan x \sec^2 x \, dx - \int \tan x \, dx \\&= \frac{\tan^2 x}{2} - \ln |\sec x| + C\end{aligned}$$

In the first integral we mentally substituted  $u = \tan x$  so that  $du = \sec^2 x \, dx$ .

If an even power of tangent appears with an odd power of secant, it is helpful to express the integrand completely in terms of  $\sec x$ . Powers of  $\sec x$  may require integration by parts, as shown in the following example.

**EXAMPLE** Find  $\int \sec^3 x \, dx$ .

**SOLUTION** Here we integrate by parts with

$$u = \sec x \qquad dv = \sec^2 x \, dx$$

$$du = \sec x \tan x \, dx \qquad v = \tan x$$

Then

$$\begin{aligned} \int \sec^3 x \, dx &= \sec x \tan x - \int \sec x \tan^2 x \, dx \\ &= \sec x \tan x - \int \sec x (\sec^2 x - 1) \, dx \\ &= \sec x \tan x - \int \sec^3 x \, dx + \int \sec x \, dx \end{aligned}$$

Using Formula 1 and solving for the required integral, we get

$$\int \sec^3 x \, dx = \frac{1}{2}(\sec x \tan x + \ln |\sec x + \tan x|) + C$$

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**EXAMPLE**

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**(Integrals involving secants and tangents)** Evaluate the following integrals:

$$(a) \int \tan^2 x \, dx, \quad (b) \int \sec^4 t \, dt, \quad \text{and} \quad (c) \int \sec^3 x \tan^3 x \, dx.$$

***Solution***

$$(a) \int \tan^2 x \, dx = \int (\sec^2 x - 1) \, dx = \tan x - x + C.$$

$$\begin{aligned} (b) \int \sec^4 t \, dt &= \int (1 + \tan^2 t) \sec^2 t \, dt && \text{Let } u = \tan t, \\ & && du = \sec^2 t \, dt. \\ &= \int (1 + u^2) \, du = u + \frac{1}{3}u^3 + C = \tan t + \frac{1}{3}\tan^3 t + C. \end{aligned}$$

$$\begin{aligned} (c) \int \sec^3 x \tan^3 x \, dx & \\ &= \int \sec^2 x (\sec^2 x - 1) \sec x \tan x \, dx && \text{Let } u = \sec x, \\ & && du = \sec x \tan x \, dx. \\ &= \int (u^4 - u^2) \, du = \frac{u^5}{5} - \frac{u^3}{3} + C = \frac{1}{5} \sec^5 x - \frac{1}{3} \sec^3 x + C. \end{aligned}$$

Integrals of the form  $\int \cot^m x \csc^n x \, dx$  can be found by similar methods because of the identity  $1 + \cot^2 x = \csc^2 x$ .

Finally, we can make use of another set of trigonometric identities:

To evaluate the integrals (a)  $\int \sin mx \cos nx \, dx$ , (b)  $\int \sin mx \sin nx \, dx$ , or (c)  $\int \cos mx \cos nx \, dx$ , use the corresponding identity:

$$(a) \quad \sin A \cos B = \frac{1}{2}[\sin(A - B) + \sin(A + B)]$$

$$(b) \quad \sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$$

$$(c) \quad \cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$$

**EXAMPLE** Evaluate  $\int \sin 4x \cos 5x \, dx$ .

**SOLUTION** This integral could be evaluated using integration by parts, but it's easier to use the identity in Equation 2(a) as follows:

$$\begin{aligned}\int \sin 4x \cos 5x \, dx &= \int \frac{1}{2}[\sin(-x) + \sin 9x] \, dx \\&= \frac{1}{2} \int (-\sin x + \sin 9x) \, dx \\&= \frac{1}{2}(\cos x - \frac{1}{9} \cos 9x) + C\end{aligned}$$

$$\int \sin 3x \cos 5x \, dx.$$

**Solution** From Equation (4) with  $m = 3$  and  $n = 5$ , we get

$$\begin{aligned}\int \sin 3x \cos 5x \, dx &= \frac{1}{2} \int [\sin(-2x) + \sin 8x] \, dx \\&= \frac{1}{2} \int (\sin 8x - \sin 2x) \, dx \\&= -\frac{\cos 8x}{16} + \frac{\cos 2x}{4} + C.\end{aligned}$$

In finding the area of a circle or an ellipse, an integral of the form  $\int \sqrt{a^2 - x^2} dx$  arises, where  $a > 0$ . If it were  $\int x\sqrt{a^2 - x^2} dx$ , the substitution  $u = a^2 - x^2$  would be effective but, as it stands,  $\int \sqrt{a^2 - x^2} dx$  is more difficult. If we change the variable from  $x$  to  $\theta$  by the substitution  $x = a \sin \theta$ , then the identity  $1 - \sin^2 \theta = \cos^2 \theta$  allows us to get rid of the root sign because

$$\sqrt{a^2 - x^2} = \sqrt{a^2 - a^2 \sin^2 \theta} = \sqrt{a^2(1 - \sin^2 \theta)} = \sqrt{a^2 \cos^2 \theta} = a |\cos \theta|$$

Notice the difference between the substitution  $u = a^2 - x^2$  (in which the new variable is a function of the old one) and the substitution  $x = a \sin \theta$  (the old variable is a function of the new one).

In general we can make a substitution of the form  $x = g(t)$  by using the Substitution Rule in reverse. To make our calculations simpler, we assume that  $g$  has an inverse function; that is,  $g$  is one-to-one. In this case, if we replace  $u$  by  $x$  and  $x$  by  $t$  in the Substitution Rule we obtain

$$\int f(x) dx = \int f(g(t))g'(t) dt$$

This kind of substitution is called *inverse substitution*.



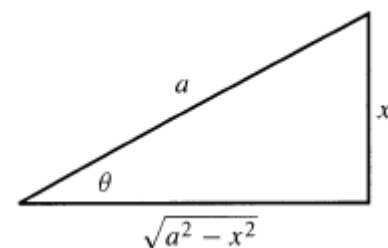
We can make the inverse substitution  $x = a \sin \theta$  provided that it defines a one-to-one function. This can be accomplished by restricting  $\theta$  to lie in the interval  $[-\pi/2, \pi/2]$ .

In the following table we list trigonometric substitutions that are effective for the given radical expressions because of the specified trigonometric identities. In each case the restric-

## The Inverse Trigonometric Substitutions

Three very useful inverse substitutions are:

$$x = a \sin \theta, \quad x = a \tan \theta, \quad \text{and} \quad x = a \sec \theta.$$



These correspond to the direct substitutions:

$$\theta = \sin^{-1} \frac{x}{a}, \quad \theta = \tan^{-1} \frac{x}{a}, \quad \text{and} \quad \theta = \sec^{-1} \frac{x}{a} = \cos^{-1} \frac{a}{x}.$$

### The inverse sine substitution

Integrals involving  $\sqrt{a^2 - x^2}$  (where  $a > 0$ ) can frequently be reduced to a simpler form by means of the substitution

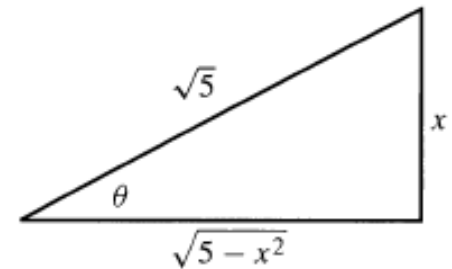
$$x = a \sin \theta \quad \text{or, equivalently,} \quad \theta = \sin^{-1} \frac{x}{a}.$$

Observe that  $\sqrt{a^2 - x^2}$  makes sense only if  $-a \leq x \leq a$ , which corresponds to  $-\pi/2 \leq \theta \leq \pi/2$ . Since  $\cos \theta \geq 0$  for such  $\theta$ , we have

$$\sqrt{a^2 - x^2} = \sqrt{a^2(1 - \sin^2 \theta)} = \sqrt{a^2 \cos^2 \theta} = a \cos \theta.$$

$$\cos \theta = \frac{\sqrt{a^2 - x^2}}{a} \quad \text{and} \quad \tan \theta = \frac{x}{\sqrt{a^2 - x^2}}.$$

Evaluate  $\int \frac{1}{(5-x^2)^{3/2}} dx$ .



$$\int \frac{1}{(5-x^2)^{3/2}} dx$$

$$\begin{aligned}\text{Let } x &= \sqrt{5} \sin \theta, \\ dx &= \sqrt{5} \cos \theta \, d\theta\end{aligned}$$

$$= \int \frac{\sqrt{5} \cos \theta \, d\theta}{5^{3/2} \cos^3 \theta}$$

$$= \frac{1}{5} \int \sec^2 \theta \, d\theta = \frac{1}{5} \tan \theta + C = \frac{1}{5} \frac{x}{\sqrt{5-x^2}} + C$$

## The inverse tangent substitution

Integrals involving  $\sqrt{a^2 + x^2}$  or  $\frac{1}{x^2 + a^2}$  (where  $a > 0$ ) are often simplified by the substitution

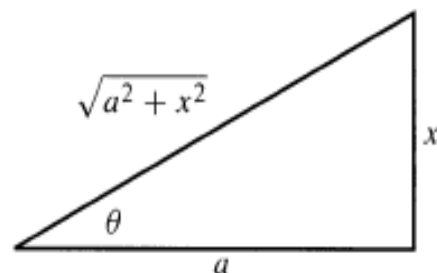
$$x = a \tan \theta \quad \text{or, equivalently,} \quad \theta = \tan^{-1} \frac{x}{a}.$$

Since  $x$  can take any real value, we have  $-\pi/2 < \theta < \pi/2$ , so  $\sec \theta > 0$  and

$$\sqrt{a^2 + x^2} = a\sqrt{1 + \tan^2 \theta} = a \sec \theta.$$

Other trigonometric functions of  $\theta$  can be expressed in terms of  $x$  by referring to a right-angled triangle with legs  $a$  and  $x$  and hypotenuse  $\sqrt{a^2 + x^2}$ .

$$\sin \theta = \frac{x}{\sqrt{a^2 + x^2}} \quad \text{and} \quad \cos \theta = \frac{a}{\sqrt{a^2 + x^2}}.$$



Evaluate (a)  $\int \frac{1}{\sqrt{4+x^2}} dx$  and (b)  $\int \frac{1}{(1+9x^2)^2} dx$ .

$$(a) \int \frac{1}{\sqrt{4+x^2}} dx$$

$$\begin{aligned} \text{Let } x &= 2 \tan \theta, \\ dx &= 2 \sec^2 \theta d\theta \end{aligned}$$

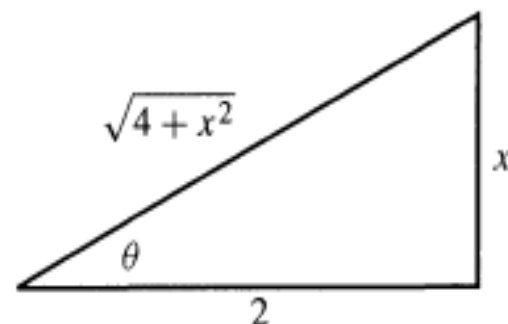
$$= \int \frac{2 \sec^2 \theta}{2 \sec \theta} d\theta$$

$$= \int \sec \theta d\theta$$

$$= \ln |\sec \theta + \tan \theta| + C = \ln \left| \frac{\sqrt{4+x^2}}{2} + \frac{x}{2} \right| + C$$

$$= \ln(\sqrt{4+x^2} + x) + C_1, \quad \text{where } C_1 = C - \ln 2.$$

(Note that  $\sqrt{4+x^2} + x > 0$  for all  $x$ , so we do not need an absolute value on it.)



$$(b) \int \frac{1}{(1+9x^2)^2} dx$$

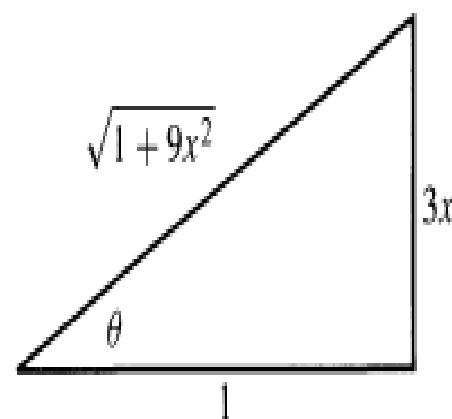
$$\begin{aligned} \text{Let } 3x &= \tan \theta, \\ 3dx &= \sec^2 \theta d\theta, \\ 1+9x^2 &= \sec^2 \theta \end{aligned}$$

$$= \frac{1}{3} \int \frac{\sec^2 \theta d\theta}{\sec^4 \theta}$$

$$= \frac{1}{3} \int \cos^2 \theta d\theta = \frac{1}{6} (\theta + \sin \theta \cos \theta) + C$$

$$= \frac{1}{6} \tan^{-1}(3x) + \frac{1}{6} \frac{3x}{\sqrt{1+9x^2}} \frac{1}{\sqrt{1+9x^2}} + C$$

$$= \frac{1}{6} \tan^{-1}(3x) + \frac{1}{2} \frac{x}{1+9x^2} + C$$



## The inverse secant substitution

Integrals involving  $\sqrt{x^2 - a^2}$  (where  $a > 0$ ) can frequently be simplified by using the substitution

$$x = a \sec \theta \quad \text{or, equivalently,} \quad \theta = \sec^{-1} \frac{x}{a}.$$

We must be more careful with this substitution. Although

$$\sqrt{x^2 - a^2} = a\sqrt{\sec^2 \theta - 1} = a\sqrt{\tan^2 \theta} = a|\tan \theta|,$$

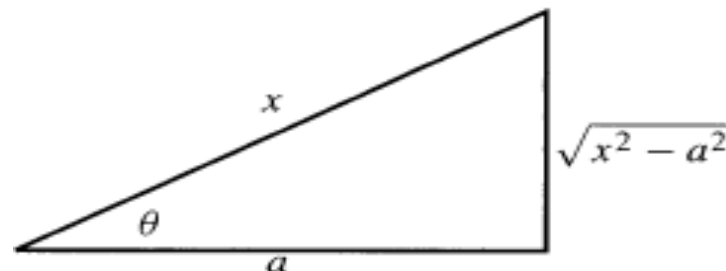
we cannot always drop the absolute value from the tangent. Observe that  $\sqrt{x^2 - a^2}$  makes sense for  $x \geq a$  and for  $x \leq -a$ .

If  $x \geq a$ , then  $0 \leq \theta = \sec^{-1} \frac{x}{a} = \arccos \frac{a}{x} < \frac{\pi}{2}$ , and  $\tan \theta \geq 0$ .

If  $x \leq -a$ , then  $\frac{\pi}{2} < \theta = \sec^{-1} \frac{x}{a} = \arccos \frac{a}{x} \leq \pi$ , and  $\tan \theta \leq 0$ .

In the first case  $\sqrt{x^2 - a^2} = a \tan \theta$ ; in the second case  $\sqrt{x^2 - a^2} = -a \tan \theta$ .

Find  $I = \int \frac{dx}{\sqrt{x^2 - a^2}}$ , where  $a > 0$ .



**Solution** For the moment, assume that  $x \geq a$ . If  $x = a \sec \theta$ , then  $dx = a \sec \theta \tan \theta d\theta$  and  $\sqrt{x^2 - a^2} = a \tan \theta$ . Thus,

$$\begin{aligned} I &= \int \sec \theta d\theta = \ln |\sec \theta + \tan \theta| + C \\ &= \ln \left| \frac{x}{a} + \frac{\sqrt{x^2 - a^2}}{a} \right| + C = \ln |x + \sqrt{x^2 - a^2}| + C_1, \end{aligned}$$

where  $C_1 = C - \ln a$ . If  $x \leq -a$ , let  $u = -x$  so that  $u \geq a$  and  $du = -dx$ . We have

$$\begin{aligned} I &= - \int \frac{du}{\sqrt{u^2 - a^2}} = - \ln |u + \sqrt{u^2 - a^2}| + C_1 \\ &= \ln \left| \frac{1}{-x + \sqrt{x^2 - a^2}} \frac{x + \sqrt{x^2 - a^2}}{x + \sqrt{x^2 - a^2}} \right| + C_1 \\ &= \ln \left| \frac{x + \sqrt{x^2 - a^2}}{-a^2} \right| + C_1 = \ln |x + \sqrt{x^2 - a^2}| + C_2, \end{aligned}$$

where  $C_2 = C_1 - 2 \ln a$ . Thus, in either case, we have

$$I = \ln |x + \sqrt{x^2 - a^2}| + C.$$



Evaluate (a)  $\int \frac{1}{\sqrt{2x - x^2}} dx$  and (b)  $\int \frac{x}{4x^2 + 12x + 13} dx$ .

$$\begin{aligned} \text{(a)} \quad \int \frac{1}{\sqrt{2x - x^2}} dx &= \int \frac{dx}{\sqrt{1 - (1 - 2x + x^2)}} \\ &= \int \frac{dx}{\sqrt{1 - (x - 1)^2}} && \begin{array}{l} \text{Let } u = x - 1, \\ du = dx \end{array} \\ &= \int \frac{du}{\sqrt{1 - u^2}} \\ &= \sin^{-1} u + C = \sin^{-1}(x - 1) + C. \end{aligned}$$

$$\begin{aligned}
\text{(b)} \quad \int \frac{x}{4x^2 + 12x + 13} dx &= \int \frac{x dx}{4 \left( x^2 + 3x + \frac{9}{4} + 1 \right)} \\
&= \frac{1}{4} \int \frac{x dx}{\left( x + \frac{3}{2} \right)^2 + 1} \quad \begin{array}{l} \text{Let } u = x + (3/2), \\ du = dx, \\ x = u - (3/2) \end{array} \\
&= \frac{1}{4} \int \frac{u du}{u^2 + 1} - \frac{3}{8} \int \frac{du}{u^2 + 1} \quad \begin{array}{l} \text{In the first integral} \\ \text{let } v = u^2 + 1, \\ dv = 2u du \end{array} \\
&= \frac{1}{8} \int \frac{dv}{v} - \frac{3}{8} \tan^{-1} u \\
&= \frac{1}{8} \ln |v| - \frac{3}{8} \tan^{-1} u + C \\
&= \frac{1}{8} \ln(4x^2 + 12x + 13) - \frac{3}{8} \tan^{-1} \left( x + \frac{3}{2} \right) + C_1,
\end{aligned}$$

where  $C_1 = C - (\ln 4)/8$ .

| Expression         | Substitution                                                                                          | Identity                            |
|--------------------|-------------------------------------------------------------------------------------------------------|-------------------------------------|
| $\sqrt{a^2 - x^2}$ | $x = a \sin \theta, \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$                              | $1 - \sin^2 \theta = \cos^2 \theta$ |
| $\sqrt{a^2 + x^2}$ | $x = a \tan \theta, \quad -\frac{\pi}{2} < \theta < \frac{\pi}{2}$                                    | $1 + \tan^2 \theta = \sec^2 \theta$ |
| $\sqrt{x^2 - a^2}$ | $x = a \sec \theta, \quad 0 \leq \theta < \frac{\pi}{2} \text{ or } \pi \leq \theta < \frac{3\pi}{2}$ | $\sec^2 \theta - 1 = \tan^2 \theta$ |

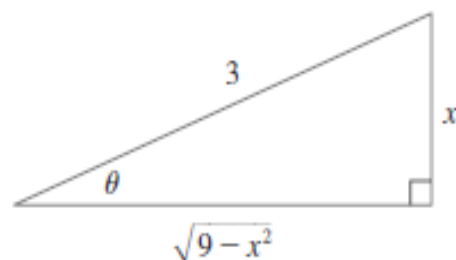
Evaluate  $\int \frac{\sqrt{9-x^2}}{x^2} dx$ .

**SOLUTION** Let  $x = 3 \sin \theta$ , where  $-\pi/2 \leq \theta \leq \pi/2$ . Then  $dx = 3 \cos \theta d\theta$  and

$$\sqrt{9-x^2} = \sqrt{9-9\sin^2\theta} = \sqrt{9\cos^2\theta} = 3|\cos\theta| = 3\cos\theta$$

(Note that  $\cos \theta \geq 0$  because  $-\pi/2 \leq \theta \leq \pi/2$ .) Thus, the Inverse Substitution Rule gives

$$\begin{aligned}\int \frac{\sqrt{9-x^2}}{x^2} dx &= \int \frac{3\cos\theta}{9\sin^2\theta} 3\cos\theta d\theta \\ &= \int \frac{\cos^2\theta}{\sin^2\theta} d\theta = \int \cot^2\theta d\theta \\ &= \int (\csc^2\theta - 1) d\theta \\ &= -\cot\theta - \theta + C\end{aligned}$$



$$\cot \theta = \frac{\sqrt{9-x^2}}{x}$$

(Although  $\theta > 0$  in the diagram, this expression for  $\cot \theta$  is valid even when  $\theta < 0$ .) Since  $\sin \theta = x/3$ , we have  $\theta = \sin^{-1}(x/3)$  and so

$$\sin \theta = \frac{x}{3}$$

$$\int \frac{\sqrt{9-x^2}}{x^2} dx = -\frac{\sqrt{9-x^2}}{x} - \sin^{-1}\left(\frac{x}{3}\right) + C$$

Find  $\int \frac{1}{x^2\sqrt{x^2+4}} dx$ .

**SOLUTION** Let  $x = 2 \tan \theta$ ,  $-\pi/2 < \theta < \pi/2$ . Then  $dx = 2 \sec^2 \theta d\theta$  and

$$\sqrt{x^2+4} = \sqrt{4(\tan^2 \theta + 1)} = \sqrt{4 \sec^2 \theta} = 2 |\sec \theta| = 2 \sec \theta$$

Thus, we have

$$\int \frac{dx}{x^2\sqrt{x^2+4}} = \int \frac{2 \sec^2 \theta d\theta}{4 \tan^2 \theta \cdot 2 \sec \theta} = \frac{1}{4} \int \frac{\sec \theta}{\tan^2 \theta} d\theta$$

To evaluate this trigonometric integral we put everything in terms of  $\sin \theta$  and  $\cos \theta$ :

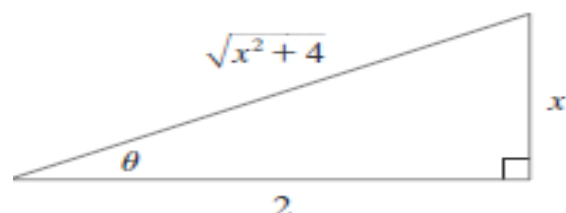
$$\frac{\sec \theta}{\tan^2 \theta} = \frac{1}{\cos \theta} \cdot \frac{\cos^2 \theta}{\sin^2 \theta} = \frac{\cos \theta}{\sin^2 \theta}$$

Therefore, making the substitution  $u = \sin \theta$ , we have

$$\begin{aligned} \int \frac{dx}{x^2\sqrt{x^2+4}} &= \frac{1}{4} \int \frac{\cos \theta}{\sin^2 \theta} d\theta = \frac{1}{4} \int \frac{du}{u^2} \\ &= \frac{1}{4} \left( -\frac{1}{u} \right) + C = -\frac{1}{4 \sin \theta} + C \\ &= -\frac{\csc \theta}{4} + C \end{aligned}$$

$$\csc \theta = \sqrt{x^2+4}/x \text{ and so}$$

$$\int \frac{dx}{x^2\sqrt{x^2+4}} = -\frac{\sqrt{x^2+4}}{4x} + C$$



$$\tan \theta = \frac{x}{2}$$

Find  $\int \frac{x}{\sqrt{x^2 + 4}} dx$ .

**SOLUTION** It would be possible to use the trigonometric substitution  $x = 2 \tan \theta$  here

But the direct substitution  $u = x^2 + 4$  is simpler, because then  $du = 2x dx$   
and

$$\int \frac{x}{\sqrt{x^2 + 4}} dx = \frac{1}{2} \int \frac{du}{\sqrt{u}} = \sqrt{u} + C = \sqrt{x^2 + 4} + C$$

## Inverse Hyperbolic Substitutions

As an alternative to the inverse secant substitution  $x = a \sec \theta$  to simplify integrals involving  $\sqrt{x^2 - a^2}$  (where  $x \geq a > 0$ ) we can use the inverse hyperbolic cosine substitution  $x = a \cosh u$ . Since  $\cosh^2 u - 1 = \sinh^2 u$ , this substitution produces  $\sqrt{x^2 - a^2} = a \sinh u$ . To express  $u$  in terms of  $x$ , we need the result,

$$\cosh^{-1} x = \ln \left( x + \sqrt{x^2 - 1} \right), \quad x \geq 1.$$

Find  $I = \int \frac{dx}{\sqrt{x^2 - a^2}}$ , where  $a > 0$ .

Again we assume  $x \geq a$ . (The case where  $x \leq -a$  can be handled similarly.) Using the substitution  $x = a \cosh u$ , so that  $dx = a \sinh u \, du$ , we have

$$\begin{aligned} I &= \int \frac{a \sinh u}{a \sinh u} du = \int du = U + C \\ &= \cosh^{-1} \frac{x}{a} + C = \ln \left( \frac{x}{a} + \sqrt{\frac{x^2}{a^2} - 1} \right) + C \\ &= \ln \left( x + \sqrt{x^2 - a^2} \right) + C_1 \quad (\text{where } C_1 = C - \ln a) \end{aligned}$$

Similarly, the inverse hyperbolic substitution  $x = a \sinh u$  can be used instead of the inverse tangent substitution  $x = a \tan \theta$  to simplify integrals involving  $\sqrt{x^2 + a^2}$  or  $\frac{1}{x^2 + a^2}$ . In this case we have  $dx = a \cosh u \, du$  and  $x^2 + a^2 = a^2 \cosh^2 u$ , and we may need the result

$$\sinh^{-1} x = \ln \left( x + \sqrt{x^2 + 1} \right)$$

valid for all  $x$



Integrals involving  $\sqrt{a^2 - x^2}$ , where  $|x| \leq a$ , can be attempted with the aid of the inverse hyperbolic substitution  $x = a \tanh u$ , making use of the identity  $1 - \tanh^2 u = \operatorname{sech}^2 u$ . However, it is usually better to use the inverse sine substitution  $x = a \sin \theta$  for such integrals.

Evaluate  $\int \frac{dx}{\sqrt{x^2 - a^2}}$ , where  $a > 0$ .

**SOLUTION 1** We let  $x = a \sec \theta$ , where  $0 < \theta < \pi/2$  or  $\pi < \theta < 3\pi/2$ . Then  $dx = a \sec \theta \tan \theta d\theta$  and

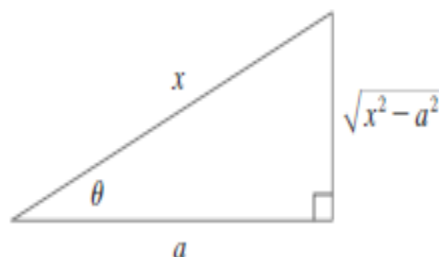
$$\sqrt{x^2 - a^2} = \sqrt{a^2(\sec^2 \theta - 1)} = \sqrt{a^2 \tan^2 \theta} = a |\tan \theta| = a \tan \theta$$

Therefore

$$\begin{aligned} \int \frac{dx}{\sqrt{x^2 - a^2}} &= \int \frac{a \sec \theta \tan \theta}{a \tan \theta} d\theta \\ &= \int \sec \theta d\theta = \ln |\sec \theta + \tan \theta| + C \end{aligned}$$

Since  $\tan \theta = \sqrt{x^2 - a^2}/a$ , so we have

$$\begin{aligned} \int \frac{dx}{\sqrt{x^2 - a^2}} &= \ln \left| \frac{x}{a} + \frac{\sqrt{x^2 - a^2}}{a} \right| + C \\ &= \ln |x + \sqrt{x^2 - a^2}| - \ln a + C \end{aligned}$$



Writing  $C_1 = C - \ln a$ , we have

$$\boxed{1} \quad \int \frac{dx}{\sqrt{x^2 - a^2}} = \ln |x + \sqrt{x^2 - a^2}| + C_1$$

**SOLUTION 2** For  $x > 0$  the hyperbolic substitution  $x = a \cosh t$  can also be used. Using the identity  $\cosh^2 y - \sinh^2 y = 1$ , we have

$$\sqrt{x^2 - a^2} = \sqrt{a^2(\cosh^2 t - 1)} = \sqrt{a^2 \sinh^2 t} = a \sinh t$$

Since  $dx = a \sinh t \, dt$ , we obtain

$$\int \frac{dx}{\sqrt{x^2 - a^2}} = \int \frac{a \sinh t \, dt}{a \sinh t} = \int dt = t + C$$

Since  $\cosh t = x/a$ , we have  $t = \cosh^{-1}(x/a)$  and

$$\boxed{2} \quad \int \frac{dx}{\sqrt{x^2 - a^2}} = \cosh^{-1}\left(\frac{x}{a}\right) + C$$

Evaluate  $\int \frac{x}{\sqrt{3-2x-x^2}} dx$ .

**SOLUTION** We can transform the integrand into a function for which trigonometric substitution is appropriate by first completing the square under the root sign:

$$\begin{aligned} 3 - 2x - x^2 &= 3 - (x^2 + 2x) = 3 + 1 - (x^2 + 2x + 1) \\ &= 4 - (x + 1)^2 \end{aligned}$$

This suggests that we make the substitution  $u = x + 1$ . Then  $du = dx$  and  $x = u - 1$ , so

$$\int \frac{x}{\sqrt{3-2x-x^2}} dx = \int \frac{u-1}{\sqrt{4-u^2}} du$$

We now substitute  $u = 2 \sin \theta$ , giving  $du = 2 \cos \theta d\theta$  and  $\sqrt{4-u^2} = 2 \cos \theta$ , so

$$\begin{aligned} \int \frac{x}{\sqrt{3-2x-x^2}} dx &= \int \frac{2 \sin \theta - 1}{2 \cos \theta} 2 \cos \theta d\theta \\ &= \int (2 \sin \theta - 1) d\theta \\ &= -2 \cos \theta - \theta + C \\ &= -\sqrt{4-u^2} - \sin^{-1}\left(\frac{u}{2}\right) + C \\ &= -\sqrt{3-2x-x^2} - \sin^{-1}\left(\frac{x+1}{2}\right) + C \end{aligned}$$

1.  $\int \frac{du}{\sqrt{a^2 + u^2}} = \sinh^{-1} \left( \frac{u}{a} \right) + C, \quad a > 0$
2.  $\int \frac{du}{\sqrt{u^2 - a^2}} = \cosh^{-1} \left( \frac{u}{a} \right) + C, \quad u > a > 0$
3.  $\int \frac{du}{a^2 - u^2} = \begin{cases} \frac{1}{a} \tanh^{-1} \left( \frac{u}{a} \right) + C, & u^2 < a^2 \\ \frac{1}{a} \coth^{-1} \left( \frac{u}{a} \right) + C, & u^2 > a^2 \end{cases}$
4.  $\int \frac{du}{u\sqrt{a^2 - u^2}} = -\frac{1}{a} \operatorname{sech}^{-1} \left( \frac{u}{a} \right) + C, \quad 0 < u < a$
5.  $\int \frac{du}{u\sqrt{a^2 + u^2}} = -\frac{1}{a} \operatorname{csch}^{-1} \left| \frac{u}{a} \right| + C, \quad u \neq 0 \text{ and } a > 0$

## The $\tan(\theta/2)$ Substitution

There is a certain special substitution that can transform an integral whose integrand is a rational function of  $\sin \theta$  and  $\cos \theta$  (i.e., a quotient of polynomials in  $\sin \theta$  and  $\cos \theta$ ) into a rational function of  $x$ . The substitution is

$$x = \tan \frac{\theta}{2} \quad \text{or, equivalently,} \quad \theta = 2 \tan^{-1} x.$$

Observe that

$$\cos^2 \frac{\theta}{2} = \frac{1}{\sec^2 \frac{\theta}{2}} = \frac{1}{1 + \tan^2 \frac{\theta}{2}} = \frac{1}{1 + x^2},$$

so

$$\begin{aligned} \cos \theta &= 2 \cos^2 \frac{\theta}{2} - 1 = \frac{2}{1 + x^2} - 1 = \frac{1 - x^2}{1 + x^2} \\ \sin \theta &= 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} = 2 \tan \frac{\theta}{2} \cos^2 \frac{\theta}{2} = \frac{2x}{1 + x^2}. \end{aligned}$$

Also,  $dx = \frac{1}{2} \sec^2 \frac{\theta}{2} d\theta$ , so

$$d\theta = 2 \cos^2 \frac{\theta}{2} dx = \frac{2 dx}{1 + x^2}.$$

In summary:

**The  $\tan(\theta/2)$  substitution**

If  $x = \tan(\theta/2)$ , then

$$\cos \theta = \frac{1 - x^2}{1 + x^2}, \quad \sin \theta = \frac{2x}{1 + x^2}, \quad \text{and} \quad d\theta = \frac{2 dx}{1 + x^2}.$$

Note that  $\cos \theta$ ,  $\sin \theta$ , and  $d\theta$  all involve only rational functions of  $x$ . <sup>1</sup>

$$\int \frac{1}{2 + \cos \theta} d\theta$$

Let  $x = \tan(\theta/2)$ , so

$$\cos \theta = \frac{1 - x^2}{1 + x^2},$$

$$d\theta = \frac{2 dx}{1 + x^2}$$

$$= \int \frac{\frac{2 dx}{1 + x^2}}{2 + \frac{1 - x^2}{1 + x^2}} = 2 \int \frac{1}{3 + x^2} dx$$

$$= \frac{2}{\sqrt{3}} \tan^{-1} \frac{x}{\sqrt{3}} + C$$

$$= \frac{2}{\sqrt{3}} \tan^{-1} \left( \frac{1}{\sqrt{3}} \tan \frac{\theta}{2} \right) + C.$$



$$\begin{aligned}
 \int \frac{dx}{1 - \cos x} &= \int \frac{1}{1 - \cos x} \cdot \frac{1 + \cos x}{1 + \cos x} dx = \int \frac{1 + \cos x}{1 - \cos^2 x} dx \\
 &= \int \frac{1 + \cos x}{\sin^2 x} dx = \int \left( \csc^2 x + \frac{\cos x}{\sin^2 x} \right) dx
 \end{aligned}$$

$$\int \frac{\tan^3 x}{\cos^3 x} dx = \int \tan^3 x \sec^3 x dx$$

$$\int \frac{\tan^3 x}{\cos^3 x} dx = \int \frac{\sin^3 x}{\cos^3 x} \frac{1}{\cos^3 x} dx = \int \frac{\sin^3 x}{\cos^6 x} dx$$

then we could have continued as follows with the substitution  $u = \cos x$ :

$$\begin{aligned} \int \frac{\sin^3 x}{\cos^6 x} dx &= \int \frac{1 - \cos^2 x}{\cos^6 x} \sin x dx = \int \frac{1 - u^2}{u^6} (-du) \\ &= \int \frac{u^2 - 1}{u^6} du = \int (u^{-4} - u^{-6}) du \end{aligned}$$

$$\int e^{\sqrt{x}} dx$$

we substitute  $u = \sqrt{x}$ . Then  $x = u^2$ , so  $dx = 2u \, du$  and

$$\int e^{\sqrt{x}} dx = 2 \int u e^u du$$

The integrand is now a product of  $u$  and the transcendental function  $e^u$  so it can be integrated by parts.

$$\int \frac{dx}{x\sqrt{\ln x}}$$

We substitute  $u = \ln x$  because its differential is  $du = dx/x$ , which occurs in the integral.

$$\int \sqrt{\frac{1-x}{1+x}} dx$$

Although the rationalizing substitution

$$u = \sqrt{\frac{1-x}{1+x}}$$

works here it leads to a very complicated rational function. An easier method is to do some algebraic manipulation Multiply-  
ing numerator and denominator by  $\sqrt{1-x}$ , we have

$$\begin{aligned} \int \sqrt{\frac{1-x}{1+x}} dx &= \int \frac{1-x}{\sqrt{1-x^2}} dx \\ &= \int \frac{1}{\sqrt{1-x^2}} dx - \int \frac{x}{\sqrt{1-x^2}} dx \\ &= \sin^{-1} x + \sqrt{1-x^2} + C \end{aligned}$$

$$\text{find } \int x\sqrt{x^2 + 2x + 4} \, dx.$$

Since the table gives forms involving  $\sqrt{a^2 + x^2}$ ,  $\sqrt{a^2 - x^2}$ , and  $\sqrt{x^2 - a^2}$ , but not  $\sqrt{ax^2 + bx + c}$ , we first complete the square:

$$x^2 + 2x + 4 = (x + 1)^2 + 3$$

If we make the substitution  $u = x + 1$  (so  $x = u - 1$ ), the integrand will involve the pattern  $\sqrt{a^2 + u^2}$ :

$$\begin{aligned} \int x\sqrt{x^2 + 2x + 4} \, dx &= \int (u - 1) \sqrt{u^2 + 3} \, du \\ &= \int u\sqrt{u^2 + 3} \, du - \int \sqrt{u^2 + 3} \, du \end{aligned}$$

The first integral is evaluated using the substitution  $t = u^2 + 3$ :

$$\int u\sqrt{u^2 + 3} \, du = \frac{1}{2} \int \sqrt{t} \, dt = \frac{1}{2} \cdot \frac{2}{3} t^{3/2} = \frac{1}{3} (u^2 + 3)^{3/2}$$

For the second integral

$$a = \sqrt{3}:$$

$$\int \sqrt{u^2 + 3} \, du = \frac{u}{2} \sqrt{u^2 + 3} + \frac{3}{2} \ln(u + \sqrt{u^2 + 3})$$

Thus

$$\begin{aligned} \int x\sqrt{x^2 + 2x + 4} \, dx \\ = \frac{1}{3} (x^2 + 2x + 4)^{3/2} - \frac{x + 1}{2} \sqrt{x^2 + 2x + 4} - \frac{3}{2} \ln(x + 1 + \sqrt{x^2 + 2x + 4}) + C \end{aligned}$$

For the second part:

$$\int \sqrt{u^2+3} du = \int \sqrt{3(\tan^2 t + 1)} \cdot \sqrt{3} \sec^2 t dt = 3 \int \sec^3 t dt$$

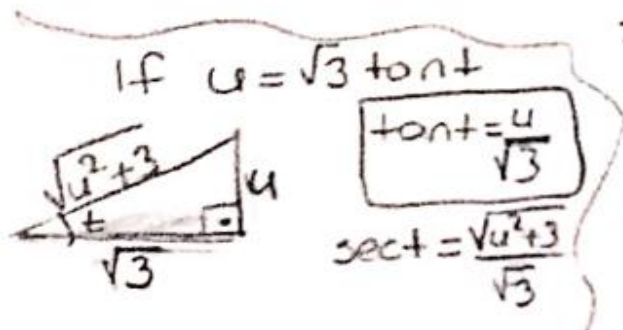
$$a=\sqrt{3} \Rightarrow u=\sqrt{3} \tan t$$

$$du = \sqrt{3} \sec^2 t dt$$

$$\tan^2 t + 1 = \sec^2 t$$

$$\Rightarrow 3 \int \sec^3 t dt = 3 \left\{ \frac{1}{2} \sec t \tan t + \frac{1}{2} \ln |\sec t + \tan t| \right\} + C$$

$$= 3 \left\{ \frac{1}{2} \frac{\sqrt{u^2+3}}{\sqrt{3}} \cdot \frac{u}{\sqrt{3}} + \frac{1}{2} \ln \left| \frac{\sqrt{u^2+3}}{\sqrt{3}} + \frac{u}{\sqrt{3}} \right| \right\} + C$$



$$= \frac{1}{2} \sqrt{u^2+3} \cdot u + \frac{3}{2} \ln \left| \frac{\sqrt{u^2+3} + u}{\sqrt{3}} \right| + C$$

$$\downarrow \sqrt{3}$$

$$\frac{3}{2} \ln |\sqrt{u^2+3} + u| - \frac{\sqrt{3}}{2} \ln 3 + C$$

$$\downarrow$$

$$C_1$$

$$= \frac{1}{2} \sqrt{u^2+3} \cdot u + \frac{3}{2} \ln |u + \sqrt{u^2+3}| + C_1$$

$$I = \int e^x \cdot \sqrt{1 - e^{2x}} \, dx = ?$$

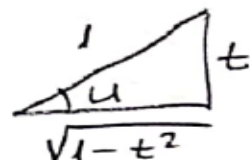
(1)

$$e^x = t \Rightarrow e^x dx = dt$$

$$I = \int \sqrt{1 - t^2} \, dt$$

$$t = \sin u$$

$$dt = \cos u \, du$$



$$= \int \sqrt{1 - \sin^2 u} \cos u \, du$$

$$= \int \cos^2 u \, du = \int \frac{1 + \cos 2u}{2} \, du$$

$$= \frac{1}{2} \left[ u + \frac{1}{2} \sin 2u \right] + C$$

$$= \frac{1}{2} \left[ u + \frac{1}{2} \cdot 2 \sin u \cos u \right] + C$$

$$= \frac{1}{2} \left[ \arcsin t + t \cdot \sqrt{1 - t^2} \right] + C$$

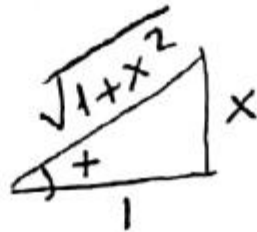
$$= \frac{1}{2} \left( \arcsin e^x + e^x \sqrt{1 - e^{2x}} \right) + C$$



$$\star \int \frac{x^2 dx}{\sqrt{(1+x^2)^3}} = ?$$

$$(2) \quad x = \tan t \Rightarrow \sec t = \sqrt{1+x^2}$$

$$dx = \sec^2 t dt$$



$$\int \frac{\tan^2 t \cdot \sec^2 t dt}{\sec^3 t} = \int \frac{\tan^2 t}{\sec t} dt$$

$$= \int \frac{\sec^2 t - 1}{\sec t} dt$$

$$= \int \sec t dt - \int \cos t dt$$

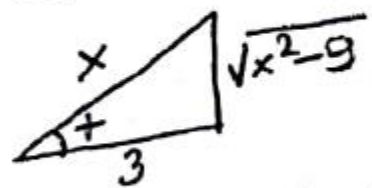
$$= \ln(\sec t + \tan t) - \sin t + C$$

$$= \ln(x + \sqrt{1+x^2}) - \frac{x}{\sqrt{1+x^2}} + C$$

$$\int \frac{9}{x^2 \sqrt{x^2-9}} dx = \int \frac{9}{\sec^2 t \cdot 3 \sec t \cdot \tan t} \cdot 3 \sec t \cdot \tan t dt$$

$$x = 3 \sec t$$

$$dx = 3 \sec t \tan t dt$$



$$= \int \cos t dt$$

$$= \sin t + C$$

$$= \frac{\sqrt{x^2-9}}{x} + C$$

$$I = \int \frac{x^2 \cdot (x^3 + 1)}{\sqrt{x^2 - 9}} dx$$

Let  $x = 3 \sec t$

$$dx = 3 \sec t \tan t dt$$

$$\frac{x}{3} = \sec t \Rightarrow \tan t = \sqrt{\frac{x^2}{9} - 1} = \frac{\sqrt{x^2 - 9}}{3}$$

$$I = \int \frac{9 \sec^2 t \cdot (27 \sec^3 t + 1)}{\sqrt{9 \sec^2 t - 9}} \cdot 3 \sec t \tan t dt$$

$$I = 243 \underbrace{\int \sec^6 t dt}_{I_1} + 9 \underbrace{\int \sec^3 t dt}_{I_2} = 243 I_1 + 9 I_2 + C$$

$$I_1 = \int \sec^6 t dt = \int (1 + \tan^2 t)^2 \cdot \sec^2 t dt$$

$$= \int \sec^2 t \, dt + 2 \int \tan^2 t \cdot \sec^2 t \, dt + \int \tan^4 t \cdot \sec^2 t \, dt$$

$$I_1 = \tan t + \frac{2}{3} \tan^3 t + \frac{1}{5} \tan^5 t$$

$$I_2 = \int \sec^3 t \, dt = \int \sec t \cdot \sec^2 t \, dt$$

$$\begin{aligned} \sec t &= u \\ \sec t \cdot \tan t \, dt &= du \end{aligned}$$

$$\begin{aligned} \sec^2 t \, dt &= dv \\ \tan t &= v \end{aligned}$$

$$= \sec t \cdot \tan t - \int \tan^2 t \cdot \sec t \, dt$$

$$I_2 = \sec t \cdot \tan t - \underbrace{\int \sec^3 t \, dt}_{I_2} + \int \sec t \, dt$$

$$2I_2 = \sec t \cdot \tan t + \int \sec t \, dt$$

$$I_2 = \frac{1}{2} \sec t \cdot \tan t + \frac{1}{2} \ln |\sec t + \tan t|$$

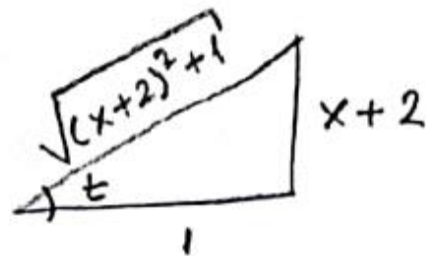
$$I = 243 \left\{ \left[ \tan t + \frac{2}{3} \tan^3 t + \frac{1}{5} \tan^5 t \right] + \frac{9}{2} \left\{ \sec t \cdot \tan t + \ln |\sec t + \tan t| \right\} \right\} + C$$

write  $\left\{ \tan t = \frac{\sqrt{x^2 - 9}}{3}, \sec t = \frac{x}{3} \right\}$  in  $I$ .

$$\int \sqrt{3x^2 + 12x + 15} \, dx = \sqrt{3} \int \sqrt{(x+2)^2 + 1} \, dx \quad (14)$$

$$x+2 = \tan t \Rightarrow dx = \sec^2 t \, dt$$

$$= \sqrt{3} \int \underbrace{\sec^3 t}_{I} \, dt$$



$$\sec t = u \quad \sec^2 t \, dt = dv$$

$$\sec t \cdot \tan t \, dt = du \quad \tan t = v$$

$$I = \left\{ \sec t \cdot \tan t - \int \tan t \cdot \sec t \cdot \tan t \, dt \right\}$$

$$I = \left\{ \sec t \cdot \tan t - \int \tan^2 t \cdot \sec t \, dt \right\}$$

An arrow points from the expression  $\sec^2 t - 1$  to the circled term  $\tan^2 t$  in the integral above.

$$I = \left\{ \text{sect.} \cdot \text{tant} - \underbrace{\int \sec^3 t dt} + \int \sec t dt \right\}$$

$$I = \frac{1}{2} \text{sect.} \cdot \text{tant} + \frac{1}{2} \ln |\sec t + \text{tant}|$$

$$\sqrt{3} \int \sec^3 dt = \sqrt{3} \left\{ \frac{1}{2} \text{sect.} \cdot \text{tant} + \frac{1}{2} \ln |\sec t + \text{tant}| \right\} + C$$

$$= \frac{\sqrt{3}}{2} \sqrt{x^2 + 4x + 5} \cdot (x+2) + \frac{\sqrt{3}}{2} \ln (\sqrt{x^2 + 4x + 5} + (x+2)) + C$$

$$I = \int \frac{dx}{(4x - x^2)^{3/2}} = ?$$

(5)

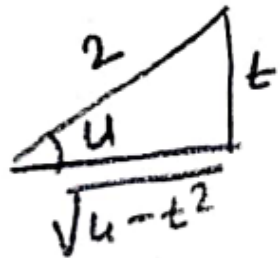
$$\begin{aligned} 4x - x^2 &= -(x^2 - 4x) = -(x^2 - 4x + 4 - 4) \\ &= -((x-2)^2 - 4) \\ &= 4 - (x-2)^2 \end{aligned}$$

$$\boxed{x-2=t \Rightarrow dx=dt}$$

$$I = \int \frac{dx}{[4 - (x-2)^2]^{3/2}} = \int \frac{dt}{[4 - t^2]^{3/2}}$$



$$\begin{aligned} t &= 2 \sin u \\ dt &= 2 \cos u \, du \end{aligned}$$



$$= \int \frac{2 \cos u \, du}{(4 - 4 \sin^2 u)^{3/2}}$$

$$= 2 \int \frac{\cos u \, du}{8 \cdot (\cos^2 u)^{3/2}}$$

$$= \frac{1}{4} \int \frac{\cos u \, du}{\cos^3 u} = \frac{1}{4} \int \sec^2 u \, du$$

$$= \frac{1}{4} \tan u + C$$

$$= \frac{1}{4} \frac{t}{\sqrt{4-t^2}} + C$$

$$= \frac{1}{4} \frac{(x-2)}{\sqrt{4-(x-2)^2}} + C$$

$$\int \frac{dx}{x \cdot \sqrt{x^2 - 3x + 2}} = ? \quad \textcircled{x = 1/t} \quad dx = -1/t^2 dt$$

(6)

$$\int \frac{-1/t^2 dt}{1/t \cdot \sqrt{1/t^2 - 3/t + 2}} = - \int \frac{1/t^2}{1/t \cdot \sqrt{1/t^2 (1 - 3t + 2t^2)}} dt$$

$$= - \int \frac{1/t^2}{1/t^2 \cdot \sqrt{1 - 3t + 2t^2}} dt$$

$$= - \frac{1}{\sqrt{2}} \int \frac{dt}{\sqrt{t^2 - \frac{3}{2}t + \frac{1}{2}}}$$

$$= - \frac{1}{\sqrt{2}} \int \frac{dt}{\sqrt{(t - \frac{3}{4})^2 - \frac{9}{16} + \frac{1}{2}}}$$

$$= -\frac{1}{\sqrt{2}} \int \frac{dt}{\sqrt{(t-\frac{3}{4})^2 - \frac{1}{16}}}$$

$$t - \frac{3}{4} = u$$

$$dt = du$$

$$= -\frac{1}{\sqrt{2}} \int \frac{du}{\sqrt{u^2 - \frac{1}{16}}}$$

$$u = \frac{1}{4} \sec v$$

$$du = \frac{1}{4} \sec v \tan v dv$$

$$= -\frac{1}{\sqrt{2}} \int \frac{\frac{1}{4} \sec v \tan v dv}{\frac{1}{4} \cdot \tan v}$$

$$= -\frac{1}{\sqrt{2}} \int \sec v dv$$

$$= -\frac{1}{\sqrt{2}} \cdot \ln(\sec v + \tan v) + C$$

$$= -\frac{1}{\sqrt{2}} \ln(4u + \sqrt{16u^2 - 1}) + C$$

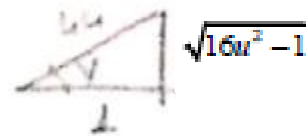
$$= -\frac{1}{\sqrt{2}} \ln(4(t - \frac{3}{4}) + \sqrt{16(t - \frac{3}{4})^2 - 1}) + C$$

$$= -\frac{1}{\sqrt{2}} \ln(4(\frac{1}{x} - \frac{3}{4}) + \sqrt{16(\frac{1}{x} - \frac{3}{4})^2 - 1}) + C$$

$$\sec^2 v - 1 = \tan^2 v$$

$$4u = \sec v = \frac{1}{\cos v}$$

$$\cos v = \frac{1}{4u}$$



$$\sec v = 4u$$

$$\frac{1}{\cos v} = 4u$$

$$\cos v = \frac{1}{4u}$$

(7)

$$\int \sqrt{1 + \cos x} \, dx = ?$$

$$u = \tan \frac{x}{2} \Rightarrow dx = \frac{2}{1+u^2} du$$

$$\int \sqrt{1 + \cos x} \, dx = 2\sqrt{2} \int \frac{du}{(u^2+1)^{3/2}}$$

$$u = \tan v \Rightarrow du = \sec^2 v \, dv$$

$$2\sqrt{2} \int \frac{du}{(u^2+1)^{3/2}} = 2\sqrt{2} \int \frac{\sec^2 v \, dv}{(\tan^2 v + 1)^{3/2}} = 2\sqrt{2} \int \frac{dv}{\sec v}$$

$$1 + \tan^2 v = \sec^2 v$$

$$\begin{aligned} \cos x &= 2\cos^2 \frac{x}{2} - 1 \\ 1 + \cos x &= 2\cos^2 \frac{x}{2} - 1 + 1 \\ 1 + \cos x &= 2\cos^2 \frac{x}{2} \\ 1 + \cos x &= 2 \cdot \left( \frac{1}{\sqrt{u^2+1}} \right)^2 = \frac{2}{u^2+1} \\ \sqrt{1 + \cos x} &= \sqrt{\frac{2}{u^2+1}} = \frac{\sqrt{2}}{\sqrt{u^2+1}} \end{aligned}$$

$$u = \tan v \Rightarrow v = 0 \text{ or } \tan u$$

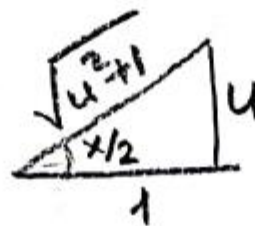
$$= 2\sqrt{2} \int \cos v dv = 2\sqrt{2} \sin v$$

$$= 2\sqrt{2} \sin(\arctan u) + C$$

$$= 2\sqrt{2} \sin(\arctan(\tan \frac{x}{2})) + C$$

$$= 2\sqrt{2} \sin \frac{x}{2} + C$$

$$u = \tan \frac{x}{2}$$



$$\Rightarrow \cos \frac{x}{2} = \frac{1}{\sqrt{u^2 + 1}}$$

$$I = \int \frac{dx}{3 + 2 \cos x} = ?$$

$$u = \tan \frac{x}{2}$$

$$\cos x = \frac{1 - u^2}{1 + u^2}$$

$$dx = \frac{2 du}{1 + u^2}$$

$$I = \int \frac{\frac{2 du}{1 + u^2}}{3 + 2 \frac{1 - u^2}{1 + u^2}}$$

$$= \int \frac{2}{\cancel{1 + u^2}} \cdot \frac{\cancel{1 + u^2}}{\underbrace{3(1 + u^2) + 2(1 - u^2)}_{3 + 3u^2 + 2 - 2u^2}} du$$

$$= 2 \int \frac{du}{u^2 + 5} = 2 \cdot \frac{1}{\sqrt{5}} \arctan \frac{u}{\sqrt{5}} + C$$

$$= \frac{2}{\sqrt{5}} \arctan \left( \frac{1}{\sqrt{5}} \cdot \tan \frac{x}{2} \right) + C$$

(8)

$$I = \int x^3 \cdot \cos x^2 dx$$

(9)

$$I = \int \underbrace{x^2}_u \cdot \underbrace{\cos x^2 \cdot x dx}_{dv}$$

$$x^2 = u$$

$$x \cdot \cos x^2 dx = dv$$

$$2x dx = du$$

$$\frac{1}{2} \sin x^2 = v$$

$$I = \frac{1}{2} \cdot x^2 \cdot \sin x^2 - \int \frac{1}{2} \sin x^2 \cdot 2 \cdot \underline{x dx}$$

$$I = \frac{1}{2} \cdot x^2 \cdot \sin x^2 - \int \sin t \cdot \frac{dt}{2}$$

$$x^2 = t$$
$$2x dx = dt$$

$$I = \frac{1}{2} \cdot x^2 \cdot \sin x^2 + \frac{1}{2} \cos t + C$$

$$I = \frac{1}{2} x^2 \cdot \sin x^2 + \frac{1}{2} \cos x^2 + C$$



$$\int \sqrt{\frac{a+x}{a-x}} dx = ?$$

$$a+x = a+a \cos 2t = a(1+\cos 2t)$$

$$= a(\cos^2 t + \cancel{\sin^2 t} + \cos^2 t - \cancel{\sin^2 t}) = 2a \cos^2 t$$

$$a-x = a-a \cos 2t = a(1-\cos 2t)$$

$$= a(\cancel{\cos^2 t} + \sin^2 t - \cancel{\cos^2 t} + \sin^2 t) = 2a \sin^2 t$$

$$\sqrt{\frac{2a \cancel{\cos^2 t}}{2a \cancel{\sin^2 t}}} = \frac{\cos t}{\sin t}$$

$$\int \frac{\cancel{\cos t}}{\cancel{\sin t}} \cdot -2a \cdot 2 \cancel{\sin t} \cos t dt = -4a \int \cos^2 t dt$$

$$\int \sqrt{\frac{a+x}{a-x}} dx = ?$$

$$x = a \cos 2t$$

$$dx = -2a \sin 2t dt$$

$$x^2 = a^2 \cos^2 2t$$

$$\cos^2 2t = \frac{x^2}{a^2}$$

$$1 - \cos^2 2t = 1 - \frac{x^2}{a^2}$$

$$\sin^2 2t = 1 - \frac{x^2}{a^2}$$

$$\sin 2t = \sqrt{1 - \frac{x^2}{a^2}}$$

$$\int \sqrt{\frac{a+x}{a-x}} dx = -4a \int \cos^2 t dt$$

$$= -4a \int \frac{1 + \cos 2t}{2} dt$$

$$= -4a \int \frac{dt}{2} - \frac{4a}{2} \int \cos 2t dt$$

$$= -2a t - 2a \cdot \frac{1}{2} \sin 2t + C$$

$$= -a \left( \arccos \frac{x}{a} + \sqrt{1 - \left( \frac{x}{a} \right)^2} \right) + C$$



## Homework

$$(1) \int \frac{\sin x}{\cos^2 x + 1} dx = ?$$

$$(2) \int \frac{\sin^3 x}{\cos x + 1} dx = ?$$

$$(3) \int \sqrt{\cos x} \cdot \sin^5 x dx = ?$$

$$(4) \int \tan^2(e^x) \cdot e^x dx = ?$$

$$(5) \int \cos(\ln x) dx = ?$$

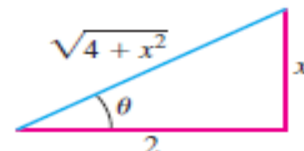
## Procedure For a Trigonometric Substitution

1. Write down the substitution for  $x$ , calculate the differential  $dx$ , and specify the selected values of  $\theta$  for the substitution.
2. Substitute the trigonometric expression and the calculated differential into the integrand, and then simplify the results algebraically.
3. Integrate the trigonometric integral, keeping in mind the restrictions on the angle  $\theta$  for reversibility.
4. Draw an appropriate reference triangle to reverse the substitution in the integration result and convert it back to the original variable  $x$ .

**EXAMPLE 1** Evaluate

$$\int \frac{dx}{\sqrt{4+x^2}}.$$

**Solution** We set



Reference triangle for  
 $x = 2 \tan \theta$  (Example 1):

$$\tan \theta = \frac{x}{2}$$

and

$$\sec \theta = \frac{\sqrt{4+x^2}}{2}.$$

$$x = 2 \tan \theta, \quad dx = 2 \sec^2 \theta \, d\theta, \quad -\frac{\pi}{2} < \theta < \frac{\pi}{2},$$

$$4 + x^2 = 4 + 4 \tan^2 \theta = 4(1 + \tan^2 \theta) = 4 \sec^2 \theta.$$

Then

$$\begin{aligned} \int \frac{dx}{\sqrt{4+x^2}} &= \int \frac{2 \sec^2 \theta \, d\theta}{\sqrt{4 \sec^2 \theta}} = \int \frac{\sec^2 \theta \, d\theta}{|\sec \theta|} \\ &= \int \sec \theta \, d\theta \\ &= \ln |\sec \theta + \tan \theta| + C \\ &= \ln \left| \frac{\sqrt{4+x^2}}{2} + \frac{x}{2} \right| + C. \end{aligned}$$

$$\sqrt{\sec^2 \theta} = |\sec \theta|$$

$$\sec \theta > 0 \text{ for } -\frac{\pi}{2} < \theta < \frac{\pi}{2}$$

**EXAMPLE 2** Evaluate

$$\int \frac{x^2 dx}{\sqrt{9 - x^2}}.$$

**Solution** We set

$$x = 3 \sin \theta, \quad dx = 3 \cos \theta d\theta, \quad -\frac{\pi}{2} < \theta < \frac{\pi}{2}$$

$$9 - x^2 = 9 - 9 \sin^2 \theta = 9(1 - \sin^2 \theta) = 9 \cos^2 \theta.$$

Then

$$\int \frac{x^2 dx}{\sqrt{9 - x^2}} = \int \frac{9 \sin^2 \theta \cdot 3 \cos \theta d\theta}{|3 \cos \theta|}$$

$$= 9 \int \sin^2 \theta d\theta$$

$$\cos \theta > 0 \text{ for } -\frac{\pi}{2} < \theta < \frac{\pi}{2}$$

$$= 9 \int \frac{1 - \cos 2\theta}{2} d\theta$$

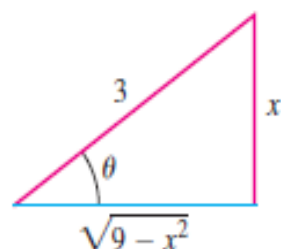
$$= \frac{9}{2} \left( \theta - \frac{\sin 2\theta}{2} \right) + C$$

$$= \frac{9}{2} (\theta - \sin \theta \cos \theta) + C$$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$= \frac{9}{2} \left( \sin^{-1} \frac{x}{3} - \frac{x}{3} \cdot \frac{\sqrt{9 - x^2}}{3} \right) + C$$

$$= \frac{9}{2} \sin^{-1} \frac{x}{3} - \frac{x}{2} \sqrt{9 - x^2} + C.$$



Reference triangle for  
 $x = 3 \sin \theta$  (Example 2):

$$\sin \theta = \frac{x}{3}$$

and

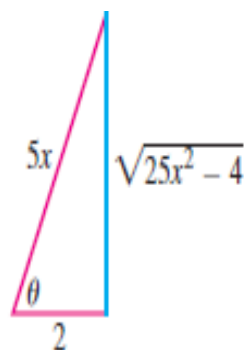
$$\cos \theta = \frac{\sqrt{9 - x^2}}{3}.$$

**EXAMPLE 3** Evaluate

$$\int \frac{dx}{\sqrt{25x^2 - 4}}, \quad x > \frac{2}{5}.$$

**Solution** We first rewrite the radical as

$$\begin{aligned}\sqrt{25x^2 - 4} &= \sqrt{25 \left( x^2 - \frac{4}{25} \right)} \\ &= 5 \sqrt{x^2 - \left( \frac{2}{5} \right)^2}\end{aligned}$$



If  $x = (2/5)\sec \theta$ ,  
 $0 < \theta < \pi/2$ , then  $\theta = \sec^{-1}(5x/2)$ ,  
 and we can read the values of the other  
 trigonometric functions of  $\theta$  from this right  
 triangle (Example 3).

to put the radicand in the form  $x^2 - a^2$ . We then substitute

$$x = \frac{2}{5} \sec \theta, \quad dx = \frac{2}{5} \sec \theta \tan \theta \, d\theta, \quad 0 < \theta < \frac{\pi}{2}$$

$$\begin{aligned}x^2 - \left( \frac{2}{5} \right)^2 &= \frac{4}{25} \sec^2 \theta - \frac{4}{25} \\ &= \frac{4}{25} (\sec^2 \theta - 1) = \frac{4}{25} \tan^2 \theta\end{aligned}$$

$$\sqrt{x^2 - \left( \frac{2}{5} \right)^2} = \frac{2}{5} |\tan \theta| = \frac{2}{5} \tan \theta.$$

$\tan \theta > 0$  for  
 $0 < \theta < \pi/2$

With these substitutions, we have

$$\begin{aligned}\int \frac{dx}{\sqrt{25x^2 - 4}} &= \int \frac{dx}{5\sqrt{x^2 - (4/25)}} = \int \frac{(2/5) \sec \theta \tan \theta d\theta}{5 \cdot (2/5) \tan \theta} \\&= \frac{1}{5} \int \sec \theta d\theta = \frac{1}{5} \ln |\sec \theta + \tan \theta| + C \\&= \frac{1}{5} \ln \left| \frac{5x}{2} + \frac{\sqrt{25x^2 - 4}}{2} \right| + C.\end{aligned}$$